

Week 1 Weekly Report

1. Work Completed

During Week 1, the following tasks were completed:

- Completed the project data requirement specification.

- Configured the Python development environment using a dedicated virtual environment.

- Configured secure API key management through a .env file.

- Successfully tested Yahoo Finance, FRED, and Alpha Vantage APIs.

- Downloaded historical market data (2018–2024), including:

 - JPM historical stock prices

 - VIX historical index values

 - U.S. Treasury Yield

- Exported the initial raw datasets for future preprocessing.

- Read Chapter 15 (Black–Scholes–Merton Model) and Chapter 26 (Exotic Options) from Hull's Options, Futures, and Other Derivatives.

- Reviewed the Advanced Chooser Option Pricing Model paper to obtain an overview of the proposed pricing framework.

2. Deliverables

Completed deliverables include:

- Data Specification Document

- API Setup Notebook

- Initial Raw Dataset (CSV files)

- Week 1 Learning Report

3. Progress Summary

All planned Week 1 objectives have been completed.

The project environment, data collection pipeline, and initial theoretical

foundation have been established. Historical financial data required for subsequent modeling has been collected and validated.

4. Issues Encountered

Required additional time to configure the Python development environment and API integration.

The mathematical derivation of the chooser option pricing model requires further study.

Further understanding of the theoretical relationship between the Black–Scholes framework and chooser option pricing will be developed in the following weeks.

5. Plan for Week 2

Perform exploratory data analysis (EDA).

Clean and preprocess the collected datasets.

Continue studying the chooser option pricing methodology.